Garage Management System (GMS) APP

The Complete Overview of Garage Management System (GMS) specification, Salesforce object setup, relationships, validations, automations, approvals, Apex triggers, PermissionSets and finally the Reports & Dashboard.

1. Project Overview

The Garage Management System (GMS) enables a garage to manage:

- Customer and vehicle details
- Service requests and workflows
- Inventory and parts usage
- Discounts, warranties, and material requests
- Notifications and approvals

- Cases via **Live web**/ Email / Phone : https://parrotcodes.github.io/GMS-Final-Project-SalesforceApp/

All data is stored and managed in Salesforce Lightning with both **standard objects** (Accounts, Vehicles(Assets), Cases, Reports, Dashboards, Operating Hours) and **custom objects** (Vehicle Services, Parts Inventory, Parts Used, Warranties, Discount Requests, Material Requests, GMS Employees, iBilling, iPayments).

2. Data Model & Relationships

Standard Objects

- **Account (Customer)** -> Stores customer details.
- **Asset (Vehicle)** -> Linked to Account, stores vehicle details.
- **Case** -> Linked to Account, Vehicle(Asset), and Vehicle Service for customer issues.

Custom Objects

- **Vehicle Service** -> Master-detail with Asset; lookup to Account and GMS Employee.
- **Parts Inventory** -> Stores garage stock.
- **Parts Used** -> Master-detail with Vehicle Service; lookup to Parts Inventory.
- **Material Request** -> Lookup to Parts Inventory and Employees.
- **Discount Request** -> Master-detail with Vehicle Service.
- **Warranty** -> Master-detail with Vehicle Service.
- **GMS Employee** -> Lookup to User; role-based (Receptionist, Worker, Supervisor, Manager, Operations).

Roll-up Summary Fields

- Vehicle Service -> SUM of Parts Used cost.
- Vehicle(Asset) -> COUNT of Vehicle Services.

3. Validation Rules

Vehicle Service

- Customer & Vehicle required.
- Status = Delivered → Payment must be Paid.
- Warranty checkbox requires expiry date.

Parts Used

- Quantity > 0.
- Quantity ≤ Current Stock.

Discount Request

- Discount Value > 0.
- Percent ≤ 50%.

Material Request

- Quantity > 0.

4. Page Layouts & Lightning Record Pages

Vehicle Service

- Sections: Customer & Vehicle, Financials, Warranty.
- Related Lists: Parts Used, Discount Requests, Cases, Warranty.
- Lightning Path: Status (New -> Assigned -> In Progress -> Done -> Delivered).
- **Quick Action :** Vehicle Service Invoice PDF***(Visual force page to download invoice)***

Parts Inventory

- Sections
- Related Lists: Parts Used, Material Requests.

5. Quick Actions

- **Request Materials** -> Prefills Material Request.
- **Submit Discount Request** -> Launches approval.

6. Process Automation (Flows)

- **Flow:** Auto Update Service Completion Date
- **Flow:** GMS_SendWelcomeEmail_On_Customer_Create

- **Flow: Vehicle** Service Discount Approval process
- **Flow:** Auto Assign Discount request to Discount Queue
- **Flow:** GMS App Material Request(Queue)
- **Flow:** Case Queue auto populate before save
- **Flow:** GMS Vehicle service auto assign to Queue

7. Approval Processes

Discount Request

- Manager approval.
- Final: Apply discount.

Material Request

- Manager approval.
- Operations approval.
- Final Status: Fulfilled

8. Apex Triggers

- GMSCreateTaskOnVehicleService (creates task after insert creating new service)
- GMSCaseQueueTrigger (when owner changed to Case assigned queue & created a Task notify to owner)
- GMSVehicleServiceTrigger (status is delivered Edit not allowed)

10. Notifications & Email Alerts

Templates:

1. Email to Assigned Employee
2. ISP Billing Status Template
3. isp payment remainder email
4. isp payment remainder email
5. isp payment remainder email
6. ISP Billing StatusTemplatae

Email Alerts:

1. Notify Assigned Employee
2. Alert - Welcome New Customer
3. Approval for discount more than 10% from manager
4. Payment Remainder for isp Bill info
5. Remainder Payment isp
6. Your Payment transaction Success

11. Reports (Garage Management System App Reports)

- New Vehicles & Fuel Type Report
- Employee Status Checking Gender Category
- New Discount Requests Report
- New Accounts with Vehicles Report
- New Leads Report
- New Cases Orgins Report

12. Dashboards (Garage Management System (GMS) app Dashboard)

- Garage Management System (GMS) app Dashboard

13. End-to-End Workflow

1. Receptionist logs customer (Account) and vehicle (Asset).
2. Create Vehicle Service (Status = New).
3. Assign Employee -> Status = Assigned -> notification sent.
4. Worker starts service -> Status = In Progress.
5. Discount Request submitted -> approvals -> discount applied.
6. Mark Done -> Status = Done –> email to customer.
7. Payment captured -> Payment Status = Paid.
8. Mark Delivered -> Status = Delivered.
9. Warranty details stored.
10. Customer issues -> Case auto-created.

14. GMS App Permission sets:

- **GMS Receptionist:**
  - Create customer records
  - Register vehicles
  - Create service requests
  - Update service status (initial, in progress)
  - Give delivery date to customer

  Objects Access:

  Accounts -> Read/Create/Edit
  Assets (Vehicles) -> Read/Create/Edit
  Vehicle Service -> Read/Create/Edit
  Parts Inventory -> Read (only)
  Discount Request -> Read (only)
  No delete access anywhere

- **GMS Worker:**
  - Work on assigned service
  - Update work progress
  - Mark service completion

  Objects Access:

  Vehicle Service -> Read/Edit
  Parts Inventory -> Read
  Parts Used -> Create/Edit
  Discount Request -> No Access
  Accounts & Vehicles -> Read-only

- **GMS Supervisor:**
  - Manage inventory
  - Create part requests
  - Approve/reject worker special requests
  - Monitor service work

- **GMS Manager:**
  - Approve Supervisor requests
  - Approve discount requests
  - Oversee garage operations

- **GMS Operation Member:**
  - Update request status
  - Notify customers
  - Manage backend process (like ordering parts)

Dummy User Examples

Name	Role	Profile & PermissionSets
Hari	Receptionist	GMS Receptionist
Ramesh Yadav	Worker	GMS Worker
Anil Kumar	Supervisor	GMS Supervisor
Sunita Reddy	Manager	GMS Manager
Vikram Singh	Operation Member	GMS Operations

Employees can now be used across our GMS app as:

Role	Responsibility
Receptionist	Creates Service record, updates customer status
Worker	Works on assigned service task
Supervisor	Manages inventory & approves special requests
Manager	Approves supervisor requests
Operations Member	Executes final operational checks

GMS app now has full employee-driven workflow:

Feature	Employee Role
Service creation	Receptionist
Work execution	Worker
Approval	Supervisor
Discount approval	Manager
Material requests	Worker/Supervisor
Inventory operations	Operations Member

Garage Management System (GMS) – Salesforce Lightning App (Live Web)

A responsive HTML & CSS website for a Garage Management System(GMS) App showcasing plans, registration, and contact forms. Includes customer registration (Web-to-Lead) and support case submission (Web-to-Case) integrated with Salesforce. Built with pure HTML5 and CSS3 for a modern, lightweight design.

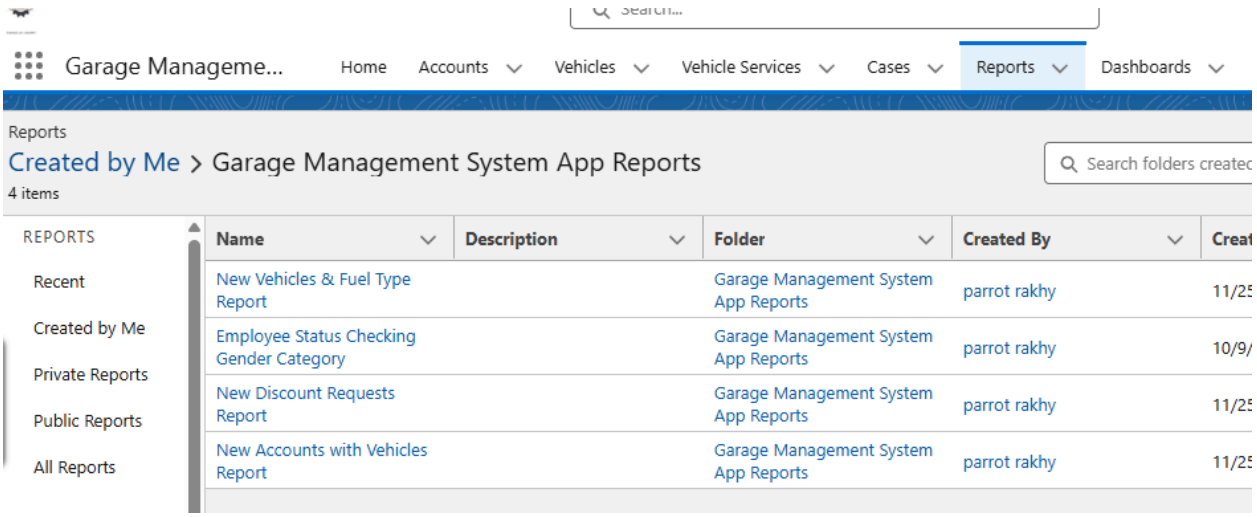
The Garage Management System (GMS) is a full-featured Salesforce Lightning application designed to streamline and automate daily operations in a modern vehicle service garage. The system is built using Salesforce Standard Objects, Custom Objects, Flows, Apex, Automation, and Lightning Web Components (LWC) to provide an end-to-end solution for managing vehicles, services, parts inventory, employees, and customer interactions.

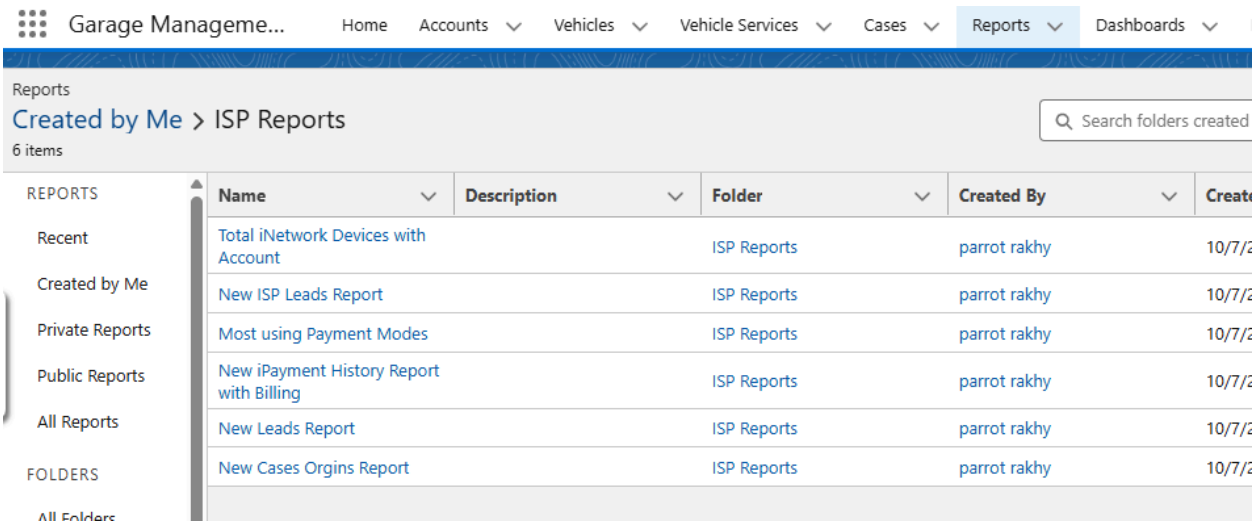
Live Web Link: https://parrotcodes.github.io/GMS-Final-Project-SalesforceApp/

Website Repo: 👉 https://github.com/Parrotcodes/GMS-Final-Project-SalesforceApp/

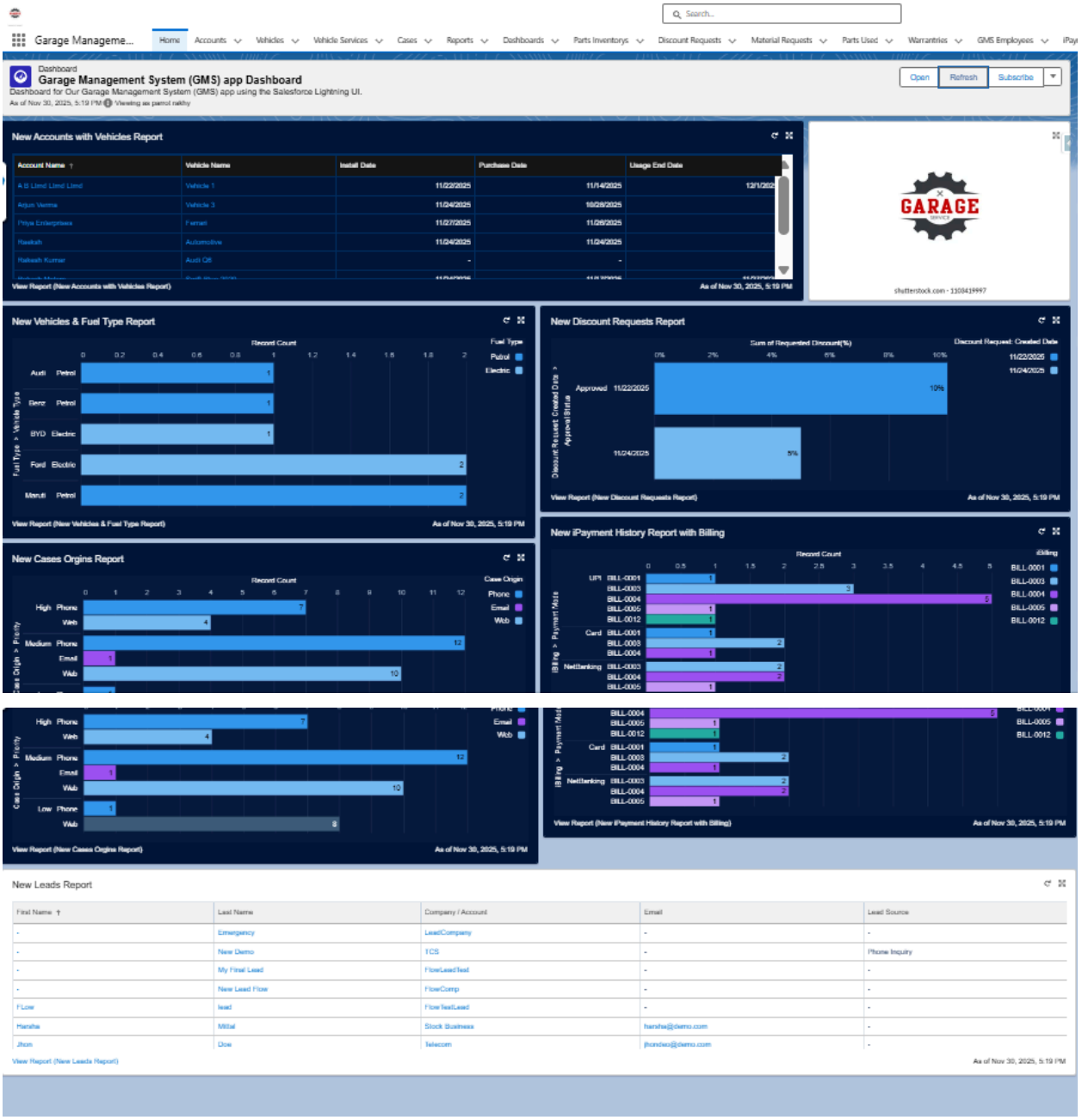
Received Cases (Salesforce Org):
https://parrotrakhy-dev-ed.develop.lightning.force.com/lightning/o/Case/list?filterName=GMS_Service_Queue_Case

Reports:





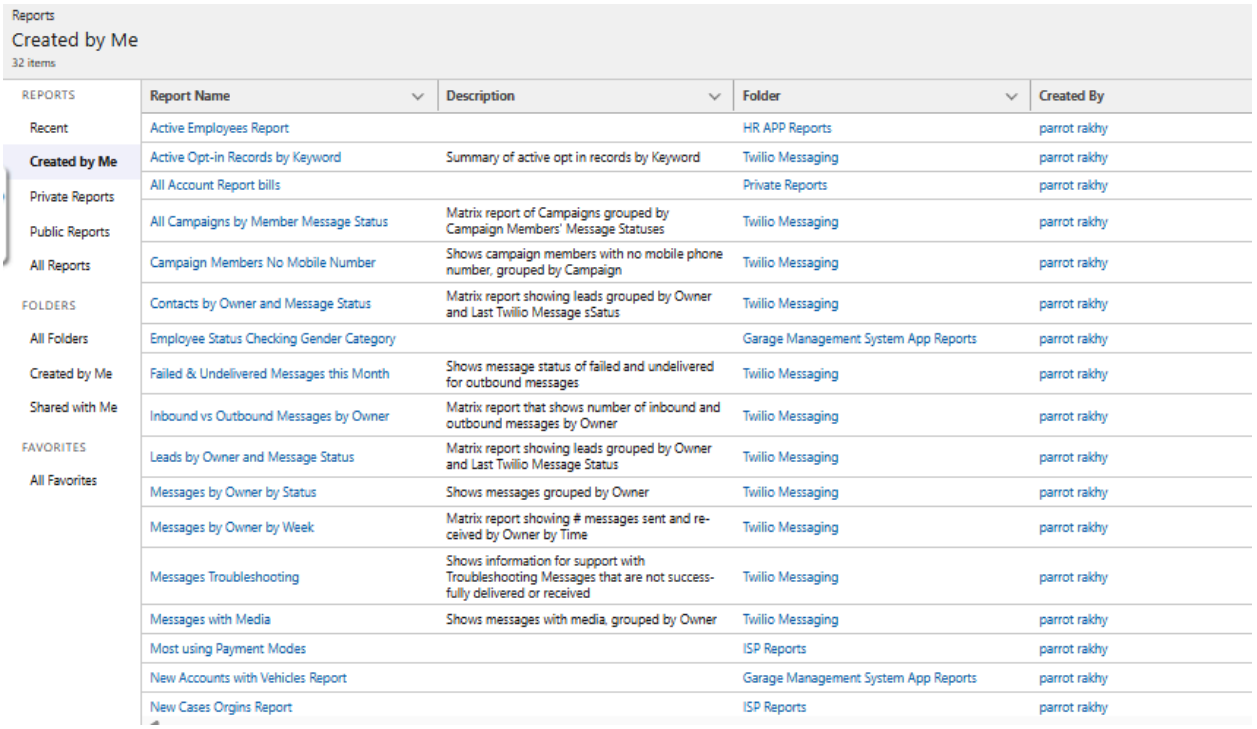
GMS App Dashboard:



All my Training Period Reports & Dashboards:

Salesforce Training (GitHub Repo) : 👉 https://github.com/Parrotcodes/Salesforce_Rakesh

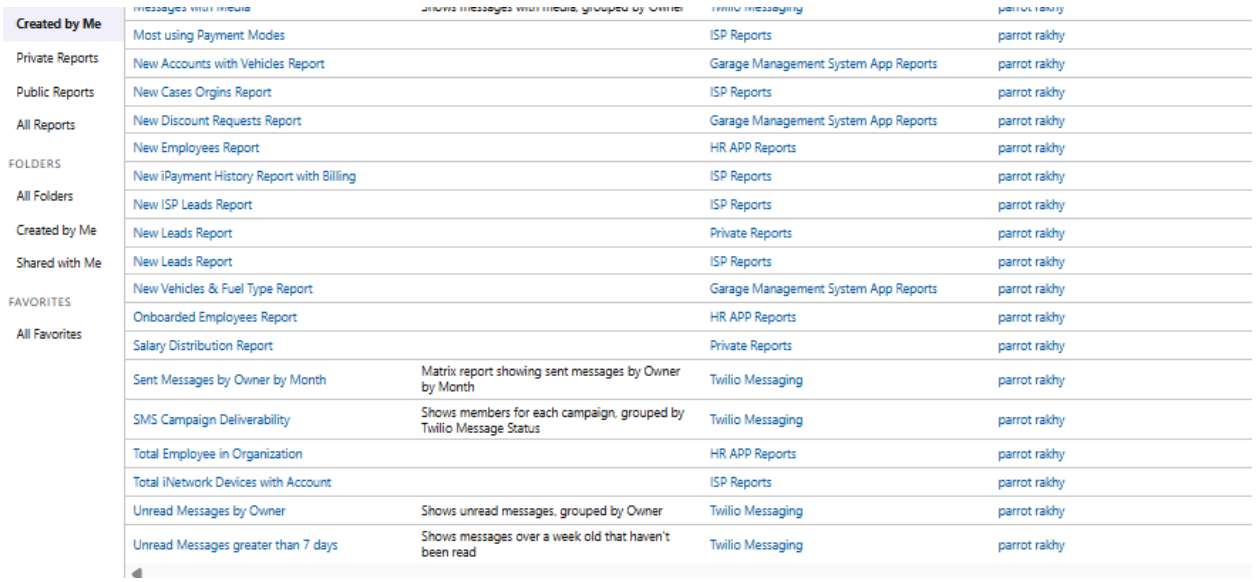
Reports:



Reports
Created by Me
32 items

REPORTS
Recent
Created by Me
Private Reports
Public Reports
All Reports
FOLDERS
All Folders
Created by Me
Shared with Me
FAVORITES
All Favorites

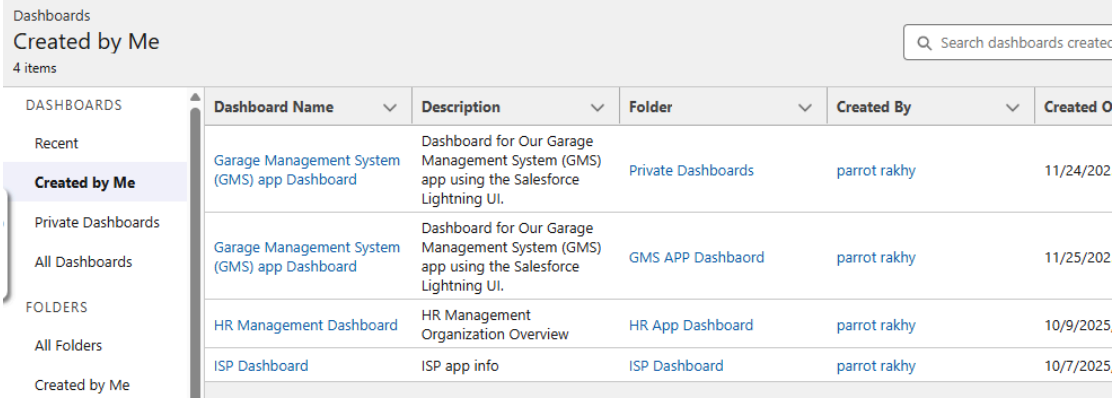
Report Name	Description	Folder	Created By
Active Employees Report		HR APP Reports	parrot rakhy
Active Opt-in Records by Keyword	Summary of active opt in records by Keyword	Twilio Messaging	parrot rakhy
All Account Report bills		Private Reports	parrot rakhy
All Campaigns by Member Message Status	Matrix report of Campaigns grouped by Campaign Members' Message Statuses	Twilio Messaging	parrot rakhy
Campaign Members No Mobile Number	Shows campaign members with no mobile phone number, grouped by Campaign	Twilio Messaging	parrot rakhy
Contacts by Owner and Message Status	Matrix report showing leads grouped by Owner and Last Twilio Message sSatus	Twilio Messaging	parrot rakhy
Employee Status Checking Gender Category		Garage Management System App Reports	parrot rakhy
Failed & Undelivered Messages this Month	Shows message status of failed and undelivered for outbound messages	Twilio Messaging	parrot rakhy
Inbound vs Outbound Messages by Owner	Matrix report that shows number of inbound and outbound messages by Owner	Twilio Messaging	parrot rakhy
Leads by Owner and Message Status	Matrix report showing leads grouped by Owner and Last Twilio Message Status	Twilio Messaging	parrot rakhy
Messages by Owner by Status	Shows messages grouped by Owner	Twilio Messaging	parrot rakhy
Messages by Owner by Week	Matrix report showing # messages sent and received by Owner by Time	Twilio Messaging	parrot rakhy
Messages Troubleshooting	Shows information for support with Troubleshooting Messages that are not successfully delivered or received	Twilio Messaging	parrot rakhy
Messages with Media	Shows messages with media, grouped by Owner	Twilio Messaging	parrot rakhy
Most using Payment Modes		ISP Reports	parrot rakhy
New Accounts with Vehicles Report		Garage Management System App Reports	parrot rakhy
New Cases Orgins Report		ISP Reports	parrot rakhy



Created by Me
Private Reports
Public Reports
All Reports
FOLDERS
All Folders
Created by Me
Shared with Me
FAVORITES
All Favorites

Most using Payment Modes		ISP Reports	parrot rakhy
New Accounts with Vehicles Report		Garage Management System App Reports	parrot rakhy
New Cases Orgins Report		ISP Reports	parrot rakhy
New Discount Requests Report		Garage Management System App Reports	parrot rakhy
New Employees Report		HR APP Reports	parrot rakhy
New iPayment History Report with Billing		ISP Reports	parrot rakhy
New ISP Leads Report		ISP Reports	parrot rakhy
New Leads Report		Private Reports	parrot rakhy
New Leads Report		ISP Reports	parrot rakhy
New Vehicles & Fuel Type Report		Garage Management System App Reports	parrot rakhy
Onboarded Employees Report		HR APP Reports	parrot rakhy
Salary Distribution Report		Private Reports	parrot rakhy
Sent Messages by Owner by Month	Matrix report showing sent messages by Owner by Month	Twilio Messaging	parrot rakhy
SMS Campaign Deliverability	Shows members for each campaign, grouped by Twilio Message Status	Twilio Messaging	parrot rakhy
Total Employee in Organization		HR APP Reports	parrot rakhy
Total iNetwork Devices with Account		ISP Reports	parrot rakhy
Unread Messages by Owner	Shows unread messages, grouped by Owner	Twilio Messaging	parrot rakhy
Unread Messages greater than 7 days	Shows messages over a week old that haven't been read	Twilio Messaging	parrot rakhy

Dashboards:



Dashboards
Created by Me
4 items
Search dashboards created

DASHBOARDS
Recent
Created by Me
Private Dashboards
All Dashboards
FOLDERS
All Folders
Created by Me

Dashboard Name	Description	Folder	Created By	Created O
Garage Management System (GMS) app Dashboard	Dashboard for Our Garage Management System (GMS) app using the Salesforce Lightning UI.	Private Dashboards	parrot rakhy	11/24/202
Garage Management System (GMS) app Dashboard	Dashboard for Our Garage Management System (GMS) app using the Salesforce Lightning UI.	GMS APP Dashbaord	parrot rakhy	11/25/202
HR Management Dashboard	HR Management Organization Overview	HR App Dashboard	parrot rakhy	10/9/2025,
ISP Dashboard	ISP app info	ISP Dashboard	parrot rakhy	10/7/2025,

All My Salesforce APPS:

